

SAR-Based Rapid Detection and GNOME Modelling of a Suspected Naphtha Oil Spill in the Northern Persian Gulf

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Abstract

Marine pollution events associated with maritime incidents require rapid assessment, particularly in regions of high oil transport such as the Persian Gulf. This study investigates a suspected oil spill following the reported attack on the Safesea Vishnu on 11 March 2026 near Basra. Due to the absence of satellite observations immediately after the incident, Sentinel-1 SAR imagery acquired on 14 March 2026 was analyzed, revealing distinct dark patches consistent with surface oil slick signatures. To reconstruct the potential evolution of the spill, a forward simulation was conducted using GNOME from 11 to 14 March 2026. The modeled results indicate a northeastward drift of the oil, with spatial patterns closely matching the dark patches observed in the SAR imagery. The use of naphtha as a representative oil type further supports the interpretation, as its rapid evaporation and spreading characteristics explain the transient and diffuse nature of the detected features. The strong spatial agreement between satellite observations and modeled dispersion patterns supports the identification of the detected anomalies as a suspected oil spill associated with the incident. This study demonstrates a novel approach in effectively integrating SAR remote sensing and numerical modeling for rapid reconstruction of marine pollution events in data-limited conditions.

Keywords: Oil spill detection; GNOME model; Oil spill trajectory; Remote sensing; Dark patch analysis; Marine pollution

Introduction

Marine oil pollution remains a critical environmental concern, particularly in regions characterized by intense maritime traffic and hydrocarbon extraction activities [1]. The Persian Gulf is one of the most important global hubs for oil production and transportation, making it

highly vulnerable to accidental spills, operational discharges, and maritime incidents [2]. Even small-scale spills can have significant ecological and economic impacts, affecting coastal ecosystems, fisheries, and navigation safety [1, 3, 2]. The semi-enclosed nature of the Persian Gulf further enhances the sensitivity of this environment to pollution events. Timely detection and monitoring of oil spills are essential for effective response and mitigation. Satellite remote sensing has become a key tool for marine surveillance, with Synthetic Aperture Radar (SAR) being particularly advantageous due to its capability for all-weather, day-and-night imaging [4, 5]. Oil films on the sea surface dampen short gravity-capillary waves, resulting in reduced radar backscatter and the appearance of dark patches in SAR imagery [3]. However, similar dark signatures may also be produced by natural phenomena such as low wind conditions, biogenic slicks, internal waves, and current shear zones, necessitating careful interpretation supported by auxiliary data [6]. On 11 March 2026, a maritime attack involving the MT Safesea Vishnu was reported near Basra (Fig. 1), in the northern sector of the Persian Gulf. Such incidents can result in the release of petroleum products due to structural damage to storage tanks or pipelines. In the present study, the spilled substance is considered to be naphtha, a light refined petroleum product with distinct physicochemical properties that significantly influence its environmental behaviour [7].

Naphtha is a low-viscosity crude oil with low surface tension, making it susceptible to spreading rapidly over the water surface, forming thin films shortly after release [8]. Its high volatility leads to rapid evaporation, often removing a substantial fraction of the spilled material within the first few hours to days after release [9]. These properties contribute to the formation of transient and spatially diffuse slicks, which may become difficult to detect in satellite imagery shortly after the initial release. Naphtha's rapid evaporation and spreading strongly influence its response to wind and currents. Under moderate winds, the remaining surface film disperses quickly, forming elongated, less concentrated slicks. This makes detection in SAR imagery highly time-sensitive, depending on how soon images are captured after the spill. Combining satellite observations with wind and current data improves understanding of slick behaviour [10, 11]. Additionally, numerical models help reconstruct spill movement and dispersion, especially when observational data are scarce.

Therefore, the objective of this study is to investigate the presence and behaviour of a suspected naphtha oil spill near Basra using an integrated approach combining satellite remote sensing, environmental data analysis, and oil spill modeling. Specifically, the study aims to identify and analyze dark patch features in satellite imagery, to evaluate prevailing wind conditions

influencing surface processes, and to simulate oil dispersion patterns using a modeling framework. This approach provides valuable information on the assessment of marine pollution events involving light petroleum products in complex and data-scarce environments.

Materials and Methods

Study Area

The study area is located in the northern part of the Persian Gulf, near the coastal region of Basra, in southern Iraq (Fig 1). This region represents a strategically important maritime zone known for its intense oil transportation and port activities. The area includes the offshore waters surrounding Khor Al Zubair Port, which serves as a major hub for oil export and ship-to-ship transfer operations. The incident involving the Safesea Vishnu occurred within this coastal marine environment on 11 March 2026. In this study, a spatial extent of approximately 30–50 km offshore from the Basra coastline was considered, encompassing the reported vessel location and the region where dark patches were detected in satellite imagery. This defined extent ensures adequate coverage for both satellite-based observation and numerical modelling of oil spill dispersion.

SAR data acquisition and preprocessing

Sentinel-1 SAR image covering the suspected spill region was downloaded and used for analysis. The downloaded image was processed in the Sentinel Application Platform (SNAP) for correcting geometric and radiometric distortions and for preparing SAR backscatter data for quantitative interpretation. The SAR image was subjected to a sequence of preprocessing steps in SNAP prior to oil spill delineation [12]. Geometric correction was first applied to correct distortions related to SAR image acquisition geometry, such as slant range effects and geolocation offsets. Radiometric correction was then performed to calibrate the SAR backscatter values so that the pixel intensities represented physically meaningful radar return values [13].

Speckle filtering and db conversion

To reduce granular noise inherent to coherent SAR imagery, speckle filtering was applied. The calibrated SAR backscatter values were converted from linear scale to decibel (dB) scale to facilitate visual interpretation and threshold-based analysis (Prasad et al. 2025). Oil slicks

appear in SAR imagery as dark patches because the presence of oil on the water surface suppresses short gravity-capillary waves, reducing radar backscatter (Alpers et al. 2017). For this study, the suspected oil spill was threshold for the local condition following Bhattacharya et al. (2026) from the range of -18 to -25 dB. Pixels within this backscatter interval were clipped and extracted as the candidate oil spill extent (Fig 2).

GNOME-based oil spill trajectory verification

The General NOAA Operational Modeling Environment (GNOME) was used to simulate the likely transport of the oil slick as a verification tool rather than for primary detection. The simulation, initialized on 11 March 2026 (the spill date), assessed whether the modeled drift and spreading patterns corresponded with the dark patch observed in the SAR image acquired on 14 March 2026. Surface current data from the HYCOM system were used to drive the model, enabling reconstruction of the probable oil movement between the release and observation period (Abdallah et al. 2022).

Results

Detection of Dark Patches

Sentinel-1 SAR imagery acquired on 14 March 2026 over the northern Persian Gulf, near Basra, revealed distinct dark patches in proximity to the reported location of the MT Safesea Vishnu incident (Fig 2). These features exhibited reduced backscatter (Fig 3) compared to the surrounding sea surface, indicating suppression of short gravity-capillary waves. The detected patches were spatially coherent, irregular in shape, and extended over several square kilometres. Their contrast and morphology are consistent with surface film signatures typically observed in SAR imagery.

Modelled Oil Spill Trajectory

An oil spill simulation was performed using GNOME, initialized at the incident location on 11 March 2026 and extended up to 14 March 2026. The model results (Fig 4) indicate a north-eastward drift of the simulated oil particles over the study period. During the initial 24 hours, rapid spreading of the oil slick was observed, followed by progressive advection and dispersion. By 14 March 2026, the simulated oil distribution covered a broader area, showing elongation along the direction of transport.

Spatial Agreement Between Observation and Model

Comparison between the Sentinel-1 SAR imagery (14 March 2026) and the modelled oil spill distribution revealed a strong spatial correspondence. The location, orientation, and extent of the simulated oil slick closely match the dark patches observed in the satellite data. This agreement indicates that the modelled transport pattern is consistent with the observed surface features.

Discussion

This study demonstrates the effectiveness of combining Sentinel-1 SAR imagery with the GNOME numerical trajectory model for reconstructing a suspected naphtha oil spill in the northern Persian Gulf. The dark patches detected in the SAR data exhibited suppressed backscatter in the range of -18 to -25 dB (Fig 3). This reduction in radar return is physically consistent with the damping of short gravity-capillary waves by surface oil films, a well-established mechanism in SAR-based marine pollution detection (Alpers et al. 2017). The use of VV polarization for spill delineation is well-supported in the literature. Studies in the Persian Gulf have consistently demonstrated that VV polarization from Sentinel-1 provides superior discrimination of oil spills compared to VH polarization (Al-Ruzouq et al. 2020).

A principal interpretive challenge in SAR-based detection is the presence of look-alikes. These are dark formations produced by low wind speeds, biogenic films, and internal waves, and they can generate backscatter signatures that resemble those of mineral oil. Natural phenomena such as low wind areas, algal blooms, and micro convective cells can produce dark patches with backscatter values similar to those of oil, generating false positives in monitoring systems (Trujillo-Acatitla et al. 2024). In this study, look-alike misclassification was mitigated through contextual corroboration. The spatial coherence and multi-kilometre extent of the detected patches, the confirmed maritime incident at the proximate location, and the strong spatial agreement with GNOME-modelled drift patterns collectively support an oil spill interpretation. This triangulated approach is consistent with operational surveillance best practices. Frameworks that integrate SAR detection with trajectory simulation have demonstrated robust performance in reducing uncertainty in spill identification (Krestenitis et al. 2019).

The applied backscatter threshold of -18 to -25 dB is consistent with published findings from comparable regional contexts. Documented Persian Gulf oil spills analyzed using Sentinel-1

have returned backscatter values within similar negative decibel ranges, with variability linked to differences in oil type, wind speed, and weathering state (Neves de Souza Júnior et al. 2023). The physicochemical properties of naphtha add further interpretive complexity. Naphtha is a highly volatile, low-viscosity petroleum product. A substantial proportion of the surface slick may evaporate within the first few hours of release (Kabil et al. 2018; Valavarasu & Sairam, 2013). The SAR image was acquired three days after the incident. The observed diffuse and spatially extended dark patches are therefore consistent with the residual signature of a substantially weathered naphtha film. Evaporation and dissolution occur in the first hours and days following a spill, progressively altering the physical and chemical properties of the oil and influencing its detectability in remote sensing imagery (Scafutto et al. 2025). This time-sensitivity of naphtha detection underscores the critical importance of rapid satellite revisit and early image acquisition in spill response operations.

The GNOME model, driven by HYCOM-derived surface currents, successfully reproduced the observed north-eastward drift of the suspected slick (Fig 4). The congruence between modelled particle distributions and satellite-detected dark patches at the 72 to 96-hour mark supports the reliability of the retrospective trajectory reconstruction. This is particularly significant given the absence of in-situ observations or near-real-time satellite coverage immediately following the incident. Recent GNOME applications in the Arabian Sea have confirmed that diffusion coefficients and wind-current forcing are critical parameters for accurately replicating observed slick positions, and that validation against SAR imagery provides a robust quality control mechanism (Bhattacharya et al. 2026). The present study extends this modelling framework to a conflict-induced spill scenario in the northern Persian Gulf. This region experiences chronic oil pollution concentrated near strategic shipping corridors, with documented monthly variations in slick occurrence and extent reaching up to 160 km² (Babagolimatikolaei, 2025).

The novelty of this work lies in several concurrent contributions. It represents one of the first SAR-based investigations of a conflict-induced naphtha spill in the Persian Gulf. The study applies naphtha-specific physicochemical parameters within GNOME to simulate a light volatile hydrocarbon under data-scarce conditions. Furthermore, it employs forward modelling as a retrospective verification strategy rather than a predictive operational tool. Integrated frameworks that combine Sentinel-1 VV polarization with ancillary modelling tools for oil spill detection in the Arabian Gulf represent an emerging methodological direction, particularly for complex multi-condition spill scenarios (Lateef et al. 2025). Future work should incorporate multi-temporal SAR acquisitions, polarimetric decomposition for improved look-alike

discrimination, and coupled weathering models to better quantify mass balance and residual oil fractions.

Conclusion

This study investigated a suspected oil spill in the northern Persian Gulf near Basra following the reported maritime attack on 11 March 2026. Sentinel-1 SAR imagery acquired on 14 March revealed distinct dark patches exhibiting characteristics consistent with surface oil slicks. Due to the absence of satellite observations immediately after the incident, a forward simulation using GNOME was performed from 11 to 14 March. The modelled north-eastward drift and spatial distribution of the simulated oil showed strong agreement with the dark patches detected in the SAR imagery. The consistency between satellite-derived features and modelled dispersion patterns supports the interpretation of the observed anomalies as a suspected oil spill associated with the incident. This study demonstrates that the integration of SAR-based detection and numerical modeling provides an effective framework for reconstructing and assessing marine pollution events in data-limited conditions. Future work should incorporate multi-temporal satellite data and in-situ observations to further validate and refine oil spill detection and modeling approaches

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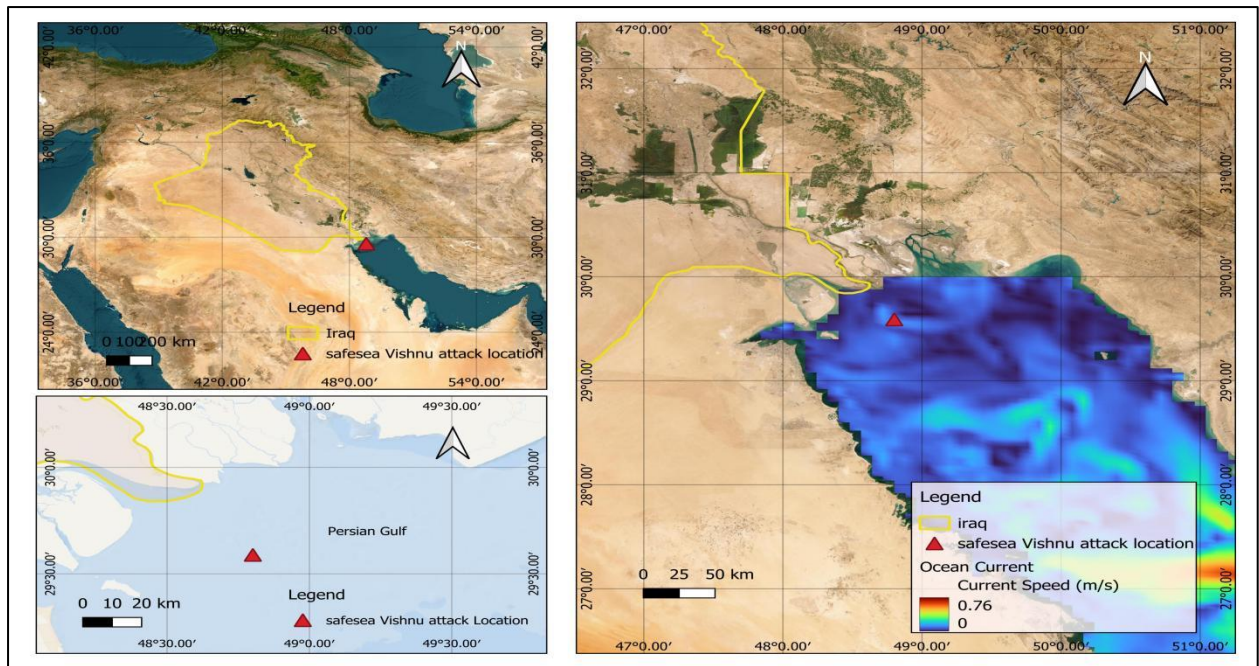


Figure 1. Geographical location of the Safesea Vishnu incident site, situated in Persian Gulf

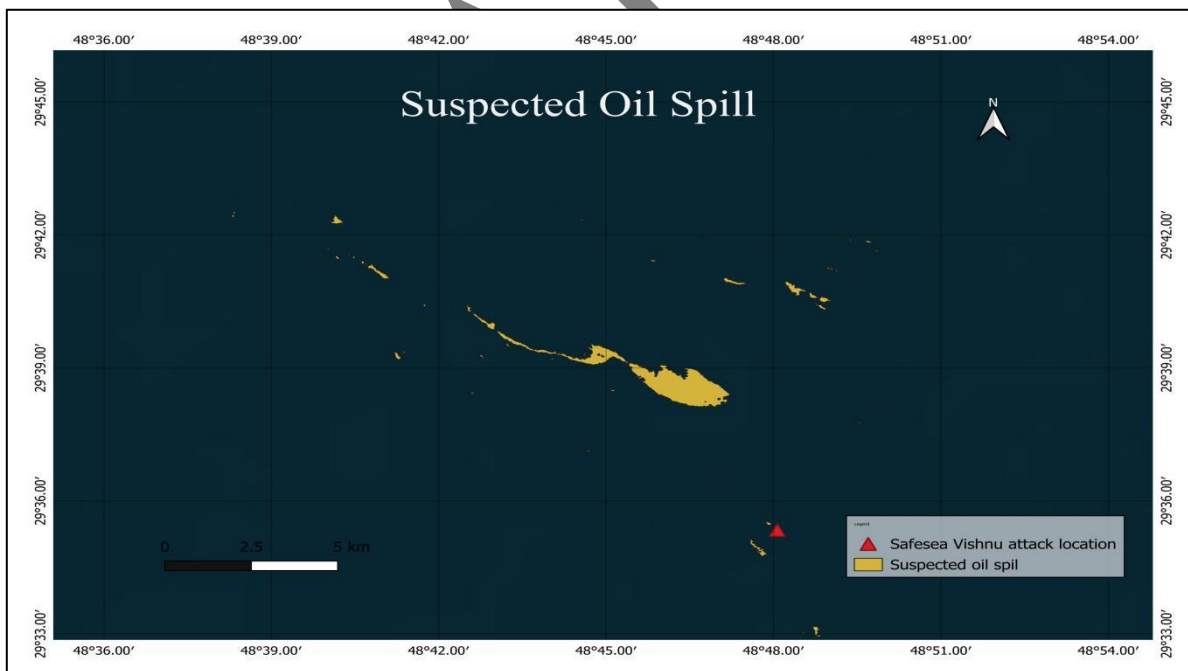


Figure 2. Map showing the suspected oil spill in Persian Gulf, near the incident location of Safesea Vishnu

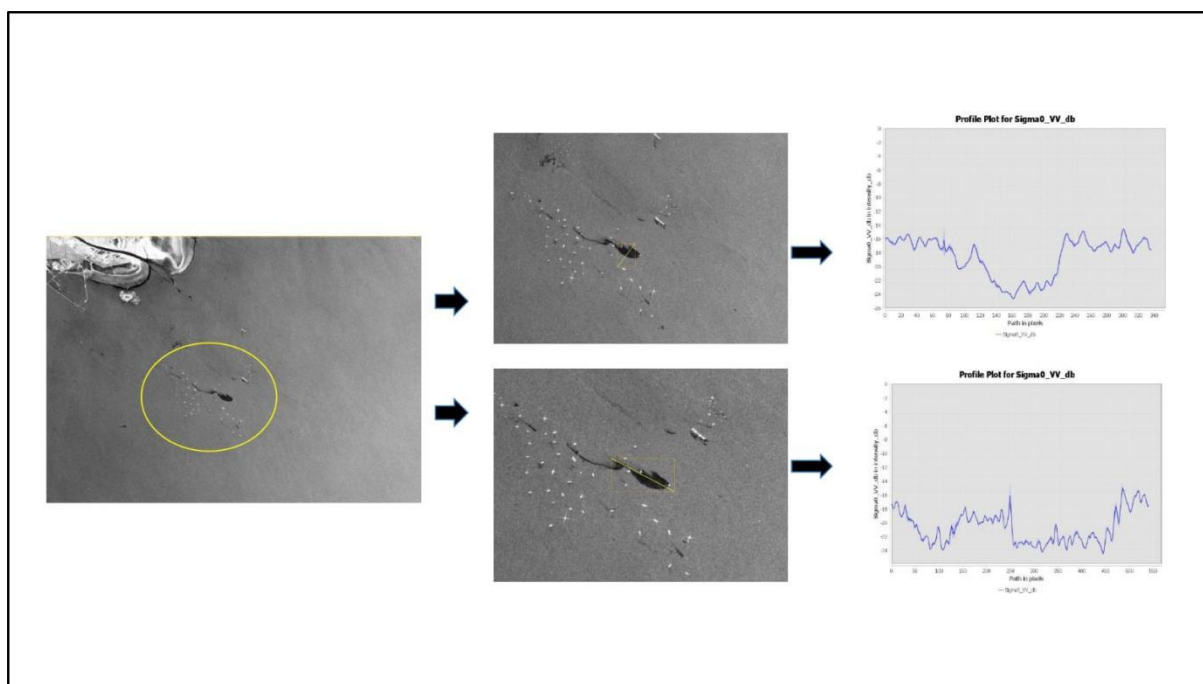


Figure 3. Visual and quantitative analysis of VV- Polarized Sentinel-1 SAR backscatter (σ^0) using SNAP, highlighting reduced σ^0 values over a potential oil spill, with sub scenes and transect-based profiles showing significant backscatter suppression across the dark patch.

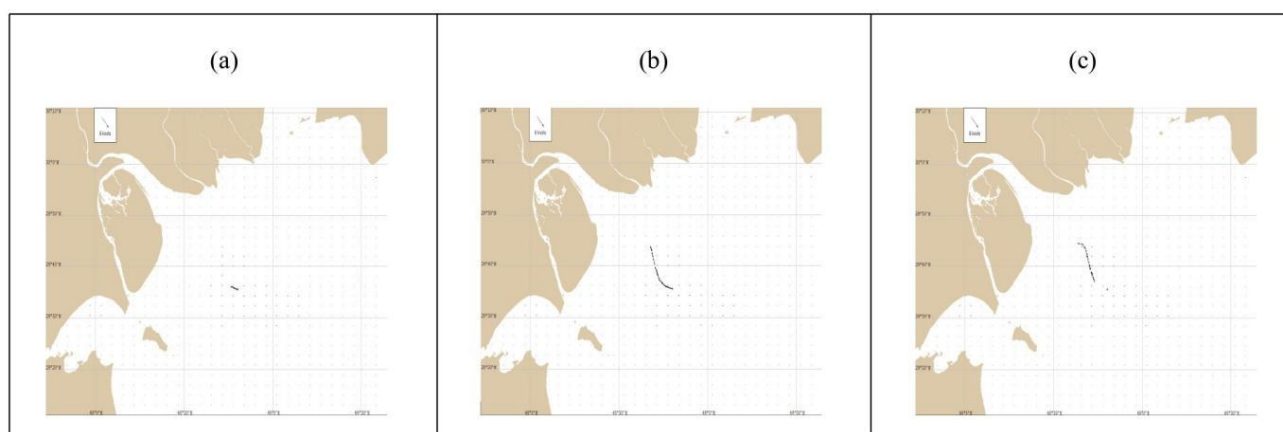


Fig. 4. GENOM simulation of oil spill drift till 14 March 2026 (a) simulation of 12.03.2026 (b) stimulation of 13.03.2026 (c) stimulation of 14.03.2026